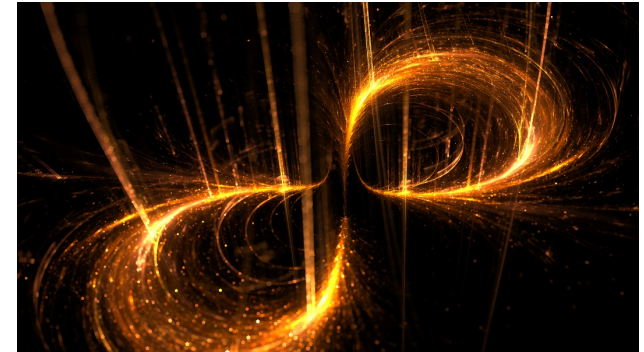


Probability

Essential Concepts

- **Experiment:** An operation which can produce some well-defined outcomes is called an experiment.



Essential Concepts

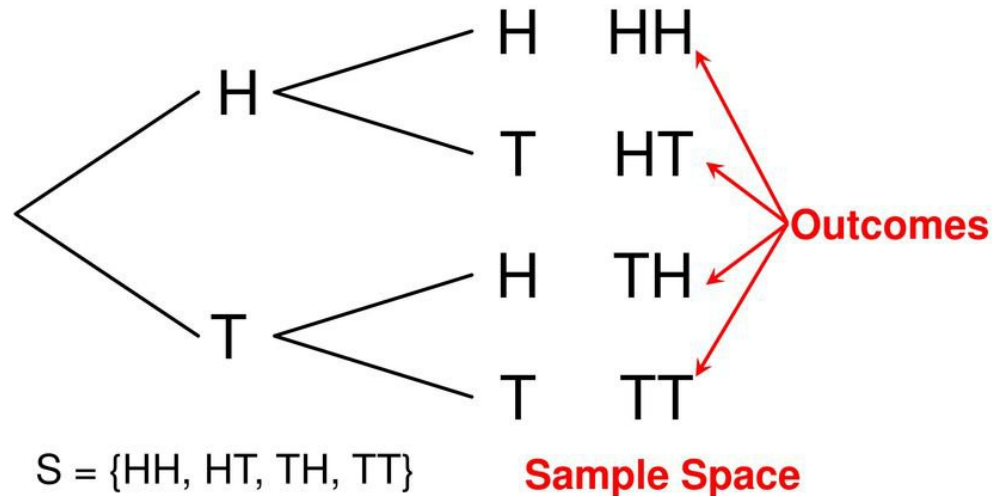
- **Random Experiment:** An experiment in which all possible outcomes are known and the exact output cannot be predicted in advance, is called a random experiment.
- **Examples:**
 - i. Rolling an unbiased dice.
 - ii. Tossing a fair coin.
 - iii. Drawing a card from a pack of well-shuffled cards.
 - iv. Picking up a ball of certain color from a bag containing balls of different colors.

How Random Experiments are Random?

1. When we throw a coin, then either a Head (H) or a Tail (T) appears.
2. A dice is a solid cube, having 6 faces, marked 1, 2, 3, 4, 5, 6 respectively. When we throw a die, the outcome is the number that appears on its upper face.
3. A pack of cards has 52 cards.
 - I. It has 13 cards of each suit, name Spades, Clubs, Hearts and Diamonds.
 - II. Cards of spades and clubs are black cards.
 - III. Cards of hearts and diamonds are red cards.
 - IV. There are 4 honors of each unit.
 - V. There are Kings, Queens and Jacks. These are all called face cards.

Essential Concepts

- **Sample Space:** When we perform an experiment, then the set S of all possible outcomes is called the sample space.
- Examples:
 - In tossing a coin, $S = \{H, T\}$
 - If two coins are tossed, the $S = \{HH, HT, TH, TT\}$.
 - In rolling a dice, we have, $S = \{1, 2, 3, 4, 5, 6\}$.



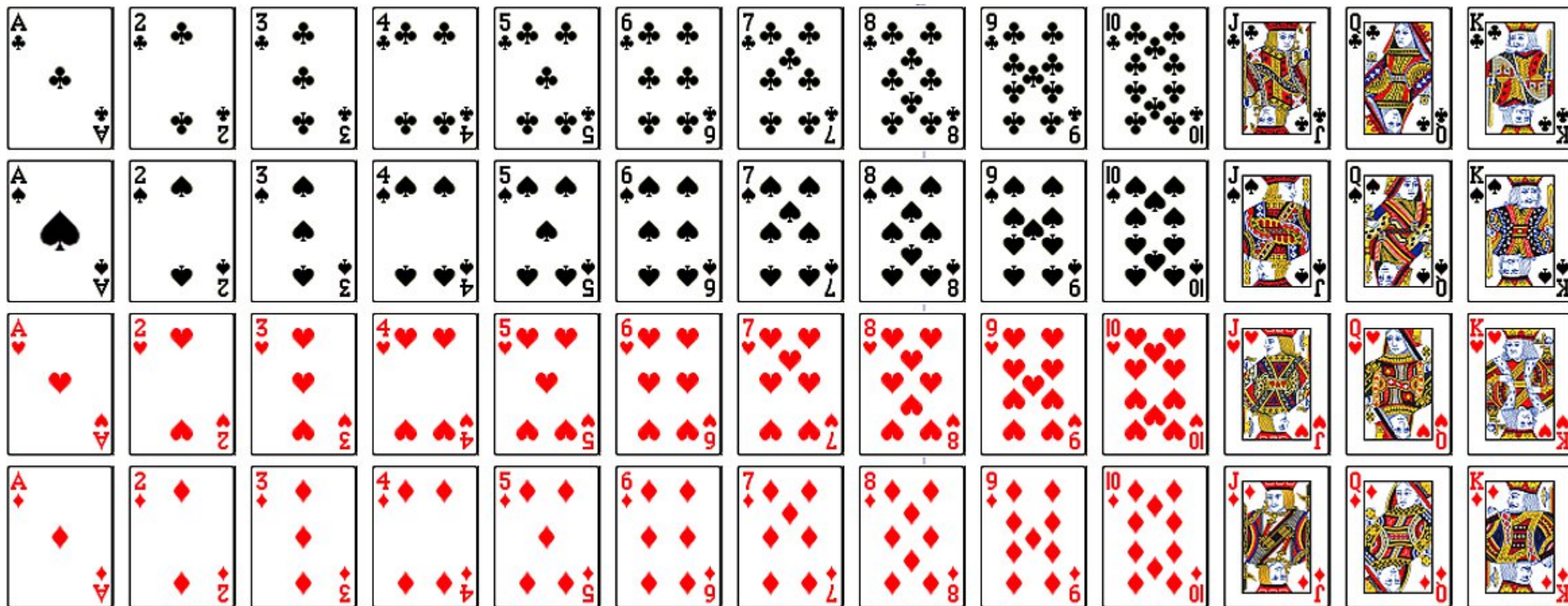
Essential Concepts

- **Event:** Any subset of a sample space is called an event.
- **Probability of Occurrence of an Event:** Let S be the sample and let E be an event.
 - Then $E \subseteq S$
 - $P(E) = \frac{n(E)}{n(S)}$
 - Here $n(E)$ is the number of elements in E and $n(s)$ is the number of elements in S .
 - The $n(X)$ is called the cardinality of X which is a fancy way of saying number of elements in X .

Essential Concepts

- Then $P(S) = 1$ (*Maximum probability is 1*)
- Range of probability $0 \leq P(E) \leq 1$
- For any events A and B, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A) + P(\bar{A}) = 1$

52 Card Deck



One standard Deck of cards



Spades ♠



A, 2, 3,
4, 5, 6,
7, 8, 9,
10,
Jack,
King,
Queen.

Hearts ♥



A, 2, 3,
4, 5, 6,
7, 8, 9,
10,
Jack,
King,
Queen.

Diamonds ♦



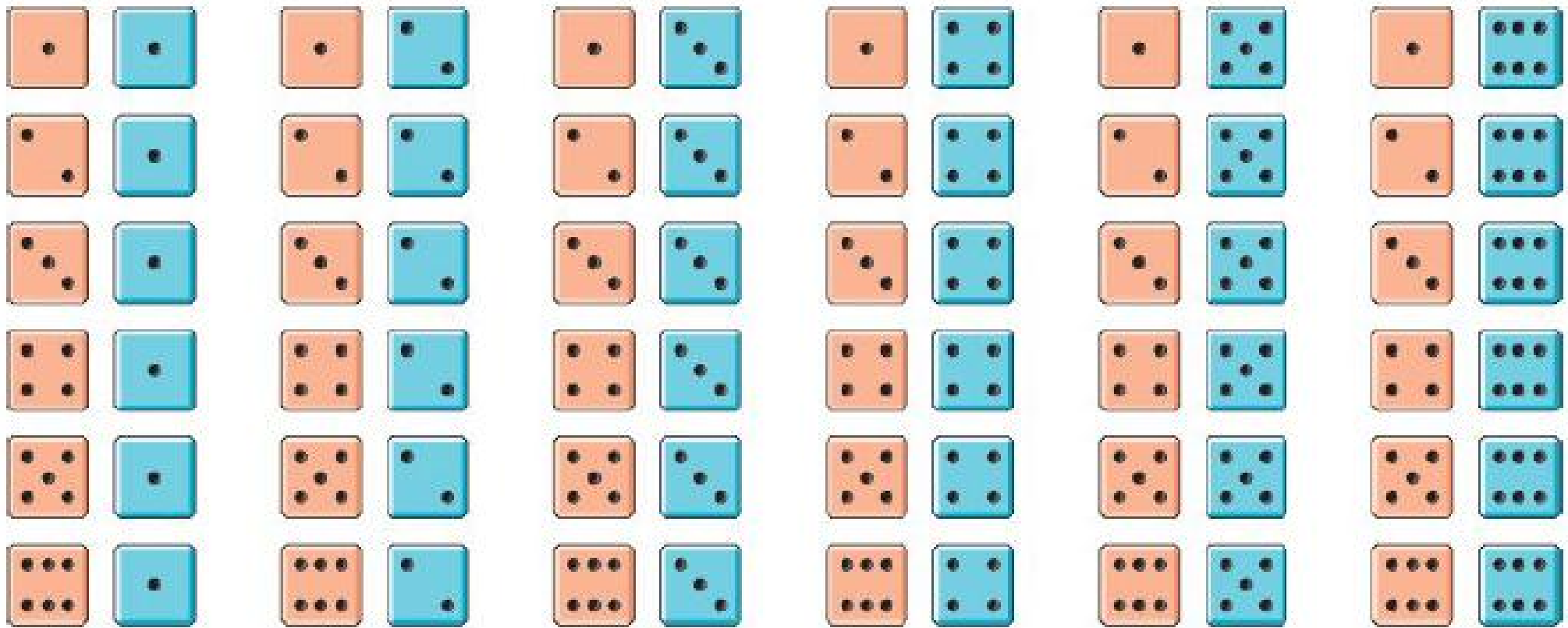
A, 2, 3,
4, 5, 6,
7, 8, 9,
10,
Jack,
King,
Queen.

Clubs ♣



A, 2, 3,
4, 5, 6,
7, 8, 9,
10,
Jack,
King,
Queen.

2 Six Sided Dice



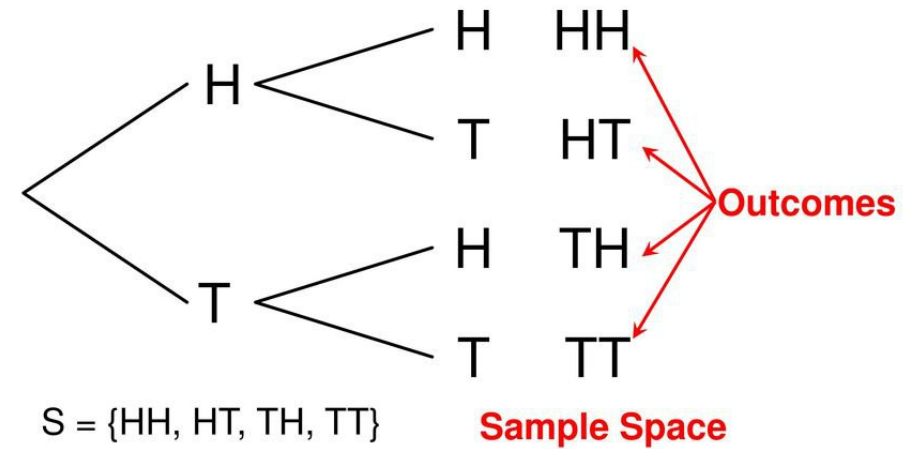
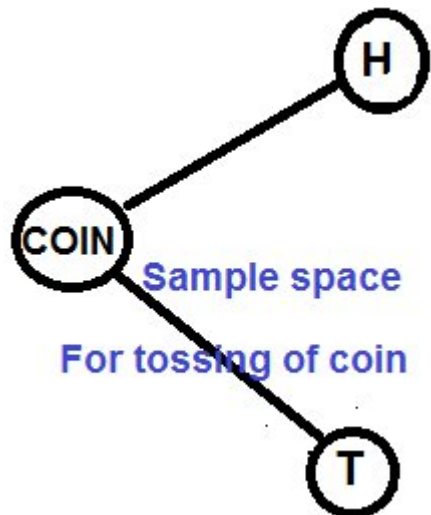
2 Six Sided Dice



	2	3	4	5	6	7
	3	4	5	6	7	8
	4	5	6	7	8	9
	5	6	7	8	9	10
	6	7	8	9	10	11
	7	8	9	10	11	12

	1	2	3	4	5	6
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

Flipping Coins



Sample Space of Flipping 2 Coins

	Heads	Tails
Heads	H, H	H, T
Tails	T, H	T, T

Sample Space: $\{HH, HT, TH, TT\}$

Exercise Problems

1. Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5? (Answer $9/20$)
2. A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue? (Answer $10/21$)
3. In a box, there are 8 red, 7 blue and 6 green balls. One ball is picked up randomly. What is the probability that it is neither red nor green? (Answer $1/3$)

Exercise Problems

1. What is the probability of getting a sum 9 from two throws of a dice? (Answer: $1/9$)
2. Three unbiased coins are tossed. What is the probability of getting at most two heads? (Answer: $7/8$)
3. Two dice are thrown simultaneously. What is the probability of getting two numbers whose product is even? (Answer: $3/4$)
4. In a class, there are 15 boys and 10 girls. Three students are selected at random. The probability that 1 girl and 2 boys are selected, is what? (Answer: $21/46$)

Exercise Problems

1. In a lottery, there are 10 prizes and 25 blanks. A lottery is drawn at random. What is the probability of getting a prize? (Answer: $2/7$)
2. From a pack of 52 cards, two cards are drawn together at random. What is the probability of both the cards being kings? (Answer: $1/221$)
3. Two dice are tossed. The probability that the total score is a prime number is what? (Answer: $5/12$)

Exercise Problems

1. A bag contains 4 white, 5 red and 6 blue balls. Three balls are drawn at random from the bag. The probability that all of them are red is what? (Answer: $2/91$)
2. Two cards are drawn together from a pack of 52 cards. The probability that one is a spade and one is a heart is what? (Answer: $13/102$)
3. One card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is a face card (Jack, Queen and King only)? (Answer: $3/13$)
4. A bag contains 6 black and 8 white balls. One ball is drawn at random. What is the probability that the ball drawn is white? (Answer: $4/7$)